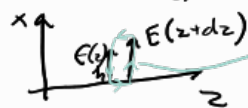


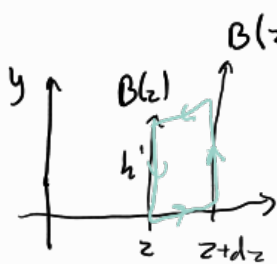
EM valovanje  
1D  $E_x, B_y, c, z$



$$\left( \int \vec{E} \cdot d\vec{s} \right) = [E_x(z+dz) - E_x(z)] \cdot \left( \frac{\partial B_y}{\partial t} dz \right)$$

$$\frac{\partial E_x}{\partial z} = - \frac{\partial B_y}{\partial t}$$

$\left| \frac{\partial}{\partial z} \right|$



$$[B_y(z+dz) - B_y(z)] \cdot \left( \frac{\partial E_x}{\partial t} dz \right) = \mu_0 \epsilon_0 \frac{\partial E_x}{\partial t} dz$$

$$\frac{\partial B_y}{\partial z} = \mu_0 \epsilon_0 \frac{\partial E_x}{\partial t}$$

$\left| \frac{\partial}{\partial t} \right|$

$$\frac{\partial^2 E_x}{\partial z^2} = - \frac{\partial^2 B_y}{\partial t^2} = \mu_0 \epsilon_0 \frac{\partial^2 E_x}{\partial t^2}$$

$$\frac{\partial E_x}{\partial z} = - \frac{\partial B_y}{\partial t}$$

$$\boxed{\frac{\partial^2 E_x}{\partial z^2} = \mu_0 \epsilon_0 \frac{\partial^2 E_x}{\partial t^2}}$$

Valovna enačba, isto za  $B_y$

$$E_x f'(z-ct) = B_y c f'(z-ct)$$

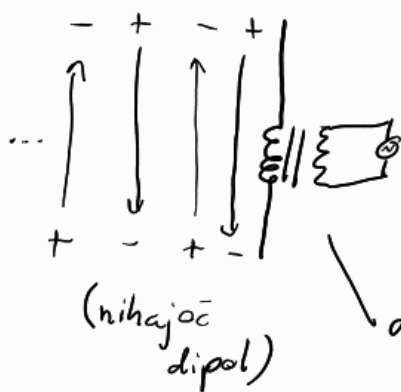
$$\boxed{E_x = c B_y}$$

$$\vec{E} = \vec{E}_x f(z \mp ct), \quad \boxed{c^2 = \frac{1}{\epsilon_0 \mu_0}}$$

( $\sin(kz \mp \omega t)$ )

$$\vec{j} = \vec{E}_x \vec{H} \left[ \frac{VA}{m^2} \right]$$

izvor:



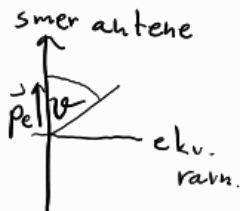
→ pospešen naboj →

$$E, B \propto \frac{1}{r}$$

$$P \propto a^2$$

$$\propto \ddot{p}_e$$

$$\propto \sin^2 \theta$$



H-atom



(razdelimo na 2 dipolni anteni)

$$P \rightarrow \frac{dW}{dt} \rightarrow \tau \sim 10^{-10} s$$

(če bi H-atom porabljal

svojo energijo za EM val.,

bi se sesedel v  $\tau$ ) → adg. dobimo iz kv. mek.

smer  $\vec{E}_0, \vec{B}_0 \rightarrow$  polarizacija

antena  $\rightarrow$  fiksna smer  $\vec{E}_0 \rightarrow$  linearno pol. valovanje  
(npr. radio, TV)

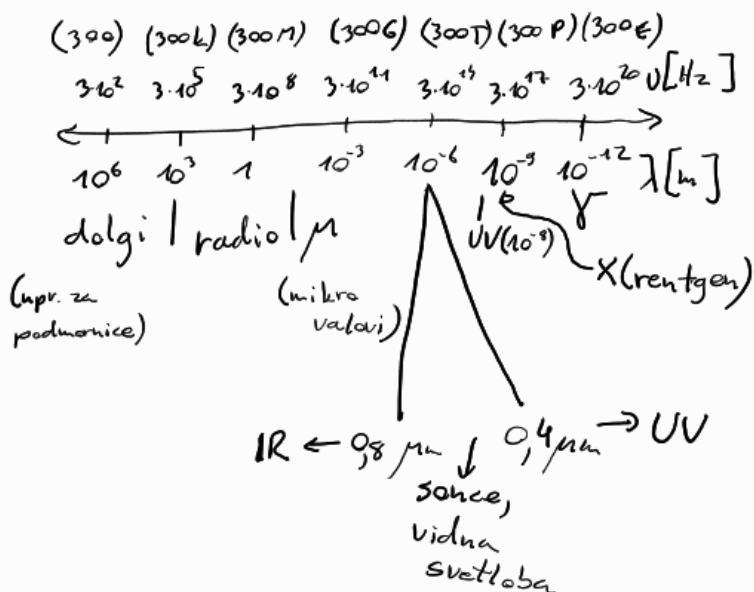
žarnica, sonce (vidna svetloba)  $\rightarrow$  krožno pol. valovanje



$$E = E_0 \cos \frac{2\pi y}{N} = E_0 \left( 1 - \frac{1}{2} \left( \frac{y}{N} \right)^2 + \dots \right)^N \xrightarrow{N \gg 1} E_0$$

$\lambda, \nu$  ?  $\rightarrow$  spekter

Maxwell  $\rightarrow c = \nu \lambda, c \neq c(\lambda)$  v vakuumu



UV, X  $\rightarrow$  izvor: atomi

$\gamma \rightarrow$  izvor: jedro

svetlobni tlak

$$F = \frac{dG}{dt} = m \nu \frac{dN}{dt}$$

$$P = \frac{dW}{dt} = \frac{m \nu^2}{2} \frac{dN}{dt}$$

$$\frac{F}{P} = \frac{2}{\nu}$$

$$j = \frac{P}{S}, p = \frac{F}{S}$$

$$\frac{p}{j} = \frac{2}{\nu} = \frac{2}{c} ?$$

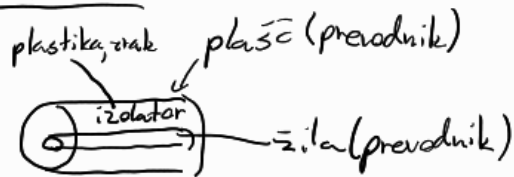
Ne velja v kl. fiziki

$$\frac{p}{j} = \frac{G}{W} = \frac{1}{c}$$

$$[p = \frac{1}{c} = w] \leftarrow \text{gest. en.}$$

$\leftarrow$  v pos. teor. relat.  $\rightarrow G = \frac{W}{c}$

# koaksialni kabel



valjast  $C \rightarrow \frac{C}{l} = \frac{2\pi\epsilon\epsilon_0}{\ln \frac{R}{r}}$

Diagram showing a segment of length  $dx$  with current  $I(x)$  entering and  $I(x+dx)$  leaving. The charge difference is  $-de = (I(x+dx) - I(x))dt$ .  

$$U = \frac{de}{dC} = \frac{de}{\frac{C}{l} dx}$$

$$\frac{\partial U}{\partial t} \cdot \frac{C}{l} = - \frac{\partial I}{\partial x}$$

$$\frac{\partial^2 I}{\partial x^2} = \frac{LC}{l^2} \frac{\partial^2 I}{\partial t^2}$$

$$\frac{1}{c^2} = \frac{L}{l} \frac{C}{l} = \frac{\mu_0 \ln \frac{R}{r}}{2\pi} \cdot \frac{2\pi\epsilon\epsilon_0}{\ln \frac{R}{r}} = \epsilon\epsilon_0\mu_0$$

$$c^2 = \frac{1}{\epsilon\epsilon_0\mu_0}$$

$$c_0 = 30 \frac{\text{cm}}{\text{ns}}$$

$$\underline{c \sim 20 \frac{\text{cm}}{\text{ns}}}$$

$$\epsilon \sim 2-3$$

$$\frac{U}{I} = Z = \frac{l\sqrt{LC}}{C l} = \sqrt{\frac{L/l}{C/l}} =$$

$$= \sqrt{\frac{\mu_0 (\ln \frac{R}{r})^2}{(2\pi)^2 \epsilon\epsilon_0}} = \frac{\ln \frac{R}{r}}{2\pi\sqrt{\epsilon}} \sqrt{\frac{\mu_0}{\epsilon_0}}$$

$$\sqrt{\frac{\mu_0}{\epsilon_0}} = 120\pi \Omega = 377\Omega \text{ upornost vakuma}$$



$$\vec{B} \perp \vec{E}$$

$$L = \frac{\Phi_m}{I} = \frac{\int \vec{B} d\vec{s}}{I} =$$

$$= \frac{\mu_0 I}{2\pi} \int_r^R \frac{l dr}{r}$$

$$\frac{L}{l} = \frac{\mu_0 \ln \frac{R}{r}}{2\pi}$$

$$U(x+dx) - U(x) = - \frac{d\Phi_m}{dt} = - dL \frac{dI}{dt} = - dx \frac{L}{l} \frac{dI}{dt}$$

$$\rightarrow \frac{\partial U}{\partial x} = - \frac{L}{l} \frac{\partial I}{\partial t}$$

$$c = \frac{1}{\sqrt{\epsilon \epsilon_0 \mu_0}}$$

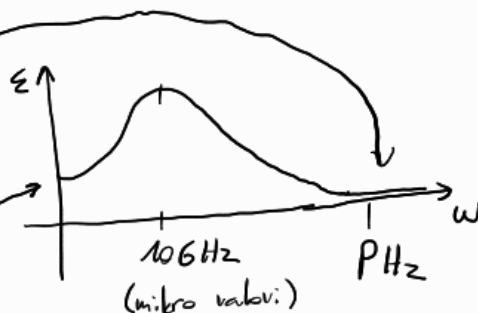
dipoli  $\rightarrow$  vsiljeno nihanje  $\rightarrow \epsilon(\omega \gg \omega_0) \ll \epsilon(\omega \sim \omega_0)$

$$n = \frac{c_0}{c} = \frac{\sqrt{\epsilon \epsilon_0 \mu_0}}{\sqrt{\epsilon_0 \mu_0}} = \sqrt{\epsilon}$$

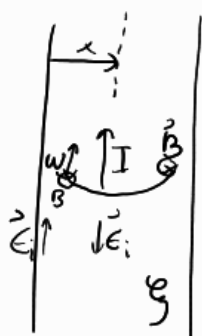
lomni  
koeficient

vdaj:  $n = \frac{4}{3} \rightarrow \epsilon(\omega \gg \omega_0) = n^2 \epsilon_0$

$$\epsilon(\omega \sim \omega_0) = 80$$



$n_{\text{rdeča}} > n_{\text{modra}}$



$$\vec{J} = \frac{1}{\eta} \vec{E}$$

$$\rightarrow E = E_0 e^{-\frac{r}{\lambda}} \rightarrow \lambda \dots \text{udornna globina}$$

$$\lambda = \sqrt{\frac{2\eta}{\mu_0 \omega}}$$

za Cu pri  $\nu = 100 \text{ MHz}$ :  $\lambda \approx 7 \cdot 10^{-6} \text{ m} \rightarrow$  tok teče skoraj samo na površju

↓  
kočni pojav (skin efekt)

$\rightarrow$  če maš visoke frekvence lahko človek prevaja el. in ne  
kumre zaradi odpovedi srca